

Instrumentation

Industrial Temperature Measurement and Control



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فروردین ماه ۱۴۰۴

- ❖ سیستمهای کنترل دمای محلی
- ❖ بیان تفاوت محلی و توزیع شده با کنترل متمرکز (در اتاق کنترل)
- ❖ نمونه های از کنترل دما محلی
- ❖ معرفی یک کارت PID کنترلر ۴ کاناله
- ❖ معرفی ترانسمیتر دما
- ❖ مزایا و هزینه استفاده از ترنسیمر
- ❖ ویژگی ها و امکانات ترانسمیترهای دما
- ❖ ترموکوپلهای صنعتی و ملاحظات
- ❖ مفاهیم IP و SIL و Zone بندی در ابزار دقیق؟
- ❖ معرفی نمونه هایی از ترنسمترهای صنعتی

❖ کاربرد سیستمهای کنترل محلی دما

○ کنترل سیستمهای کنترلی مجتمع

○ کوره های کوچک

○ کنترل توزیع شده با بردهای کنترلی

❖ در بسیاری موارد ارتباط بدون واسطه حسگر به سیستم کنترل

❖ منطق های کنترلی (on/off, PI, PID, PWM)

❖ انواع خروجی برای محرک (رله، رله حالت جامد، خروجی برای تریستور)

❖ در برخی موارد امکان مانیتورینگ و کنترل شدنِ بالا دستی برای Setpoint و

...



○ بررسی کاتالوگ راه حل کنترل دما از یک شرکت:

[Custom Heater Manufacturer | Electric Heaters | Temperature Sensors \(durexindustries.com\)](http://durexindustries.com)

C 4X 4-Channel Modular PID Controller

Standard features: Four universal process inputs including thermocouple, RTD, analog 0-10 VDC, 4-20 mA with custom linearization feature, auto-tuning PID, OFF/ ON Control, heat only, cool only, and heat/cool. Up to 8 freely assigned alarms with AND/OR Logic. Alarms can be set as absolute, deviation, direct , reverse, latching, and disabled at power up. Other features include diagnostics, auto/manual, soft-start, and standard Modbus RTU communications.

1.2 Features

- Solid state relay control
- Current transformers (four)
- 4 universal main inputs
- 4 heat/cool independent PID
- 4 main output internally wired to the SSR
- 4 auxiliary analog inputs (option)
- 4 configurable output (option): relay / logic / analog/ TRIAC
- 2 configurable relay alarm output
- 2 digital inputs
- Standard digital communication: Modbus RTU/ RS485

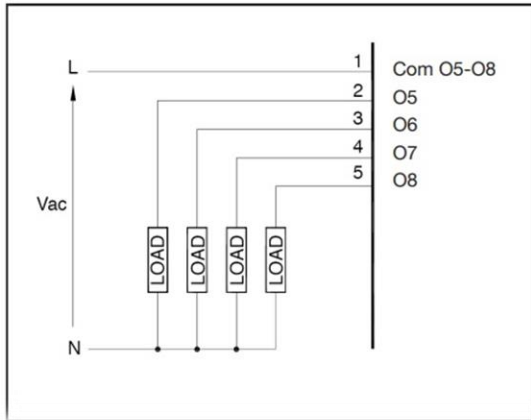


C 4x Board

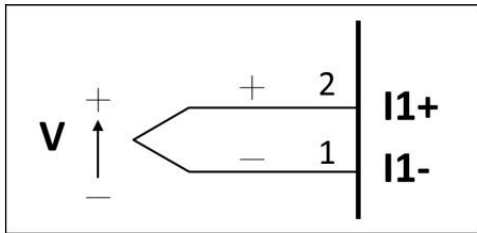
Outputs 5 - 8: TRIAC Type

TRIAC outputs: Voltage: 24...230Vac, max 1A

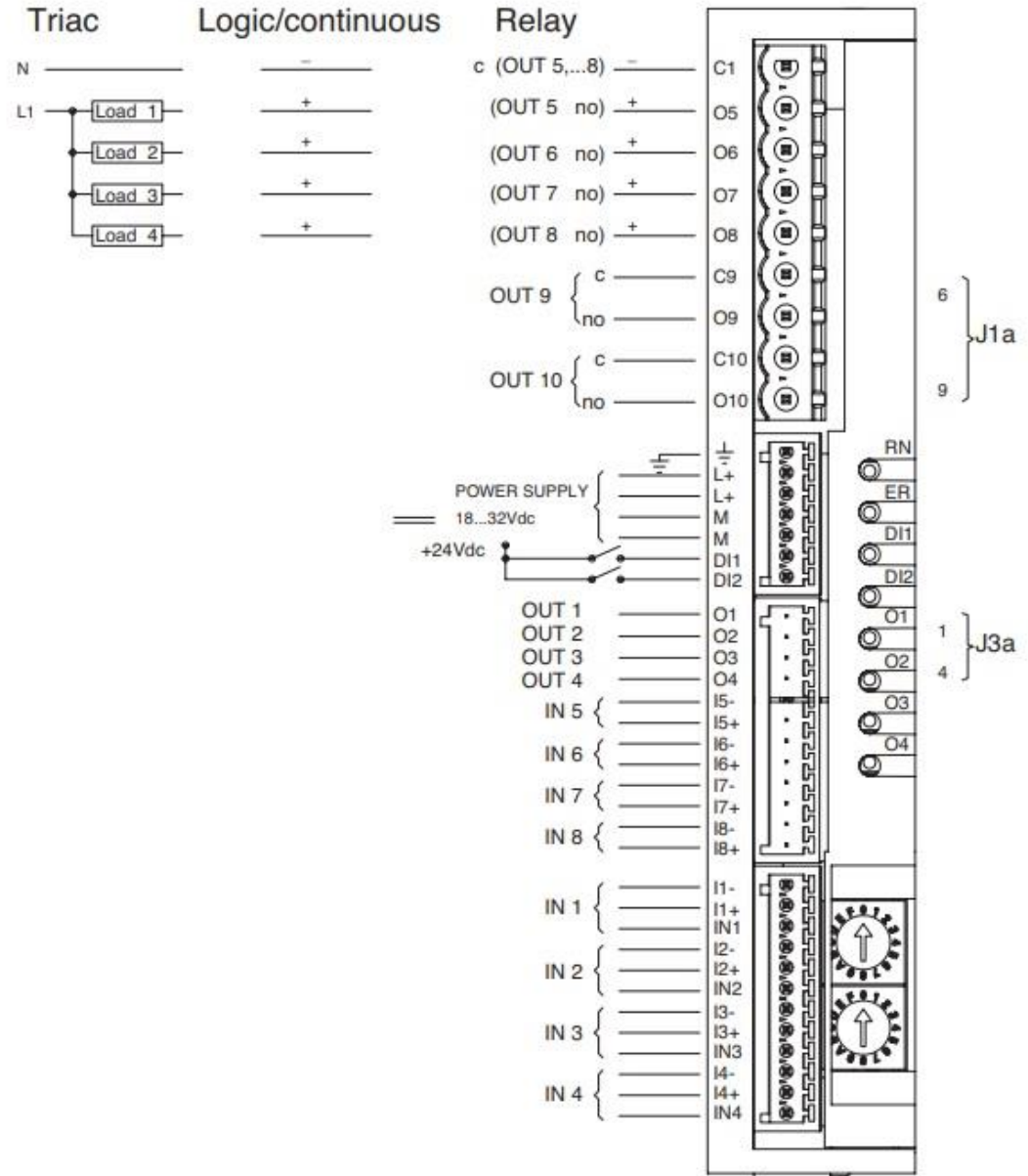
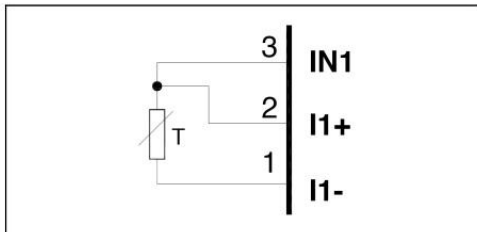
Wiring Schematic for Outputs 5 - 8, TRIAC Outputs



Wiring Schematic for 60mV TC or Linear (Analog) input

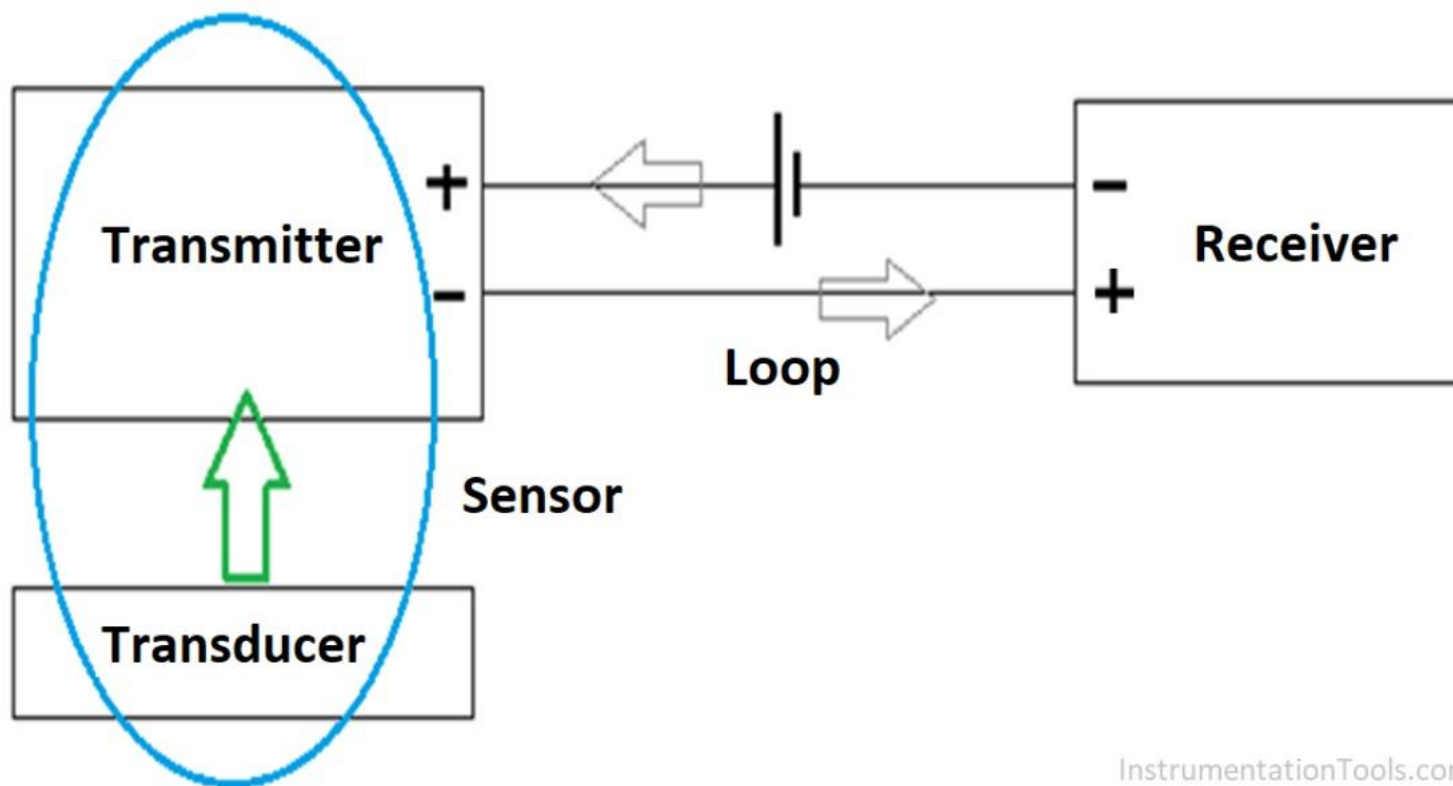


Wiring Schematic for RTD (Pt100) input



A temperature transmitter (TT) is an electrical instrument that interfaces a temperature gauge – such as a thermocouple, RTD, or thermistor – to a measurement or control device, such as a PLC, DCS, PC, loop controller, data logger, display or recorder. Typically, temperature transmitters isolate, amplify, filter noise, linearize, and convert the input signal from the sensor then send (transmit) a standardized output signal to the control device. Common electrical output signals used in manufacturing plants are 4-20mA or 0-10V DC ranges [or other protocols: HART, Fieldbus, ...] . For example, 4mA could represent 0°C and 20mA means 100°C.

Ex: Transmitter with analogue current output



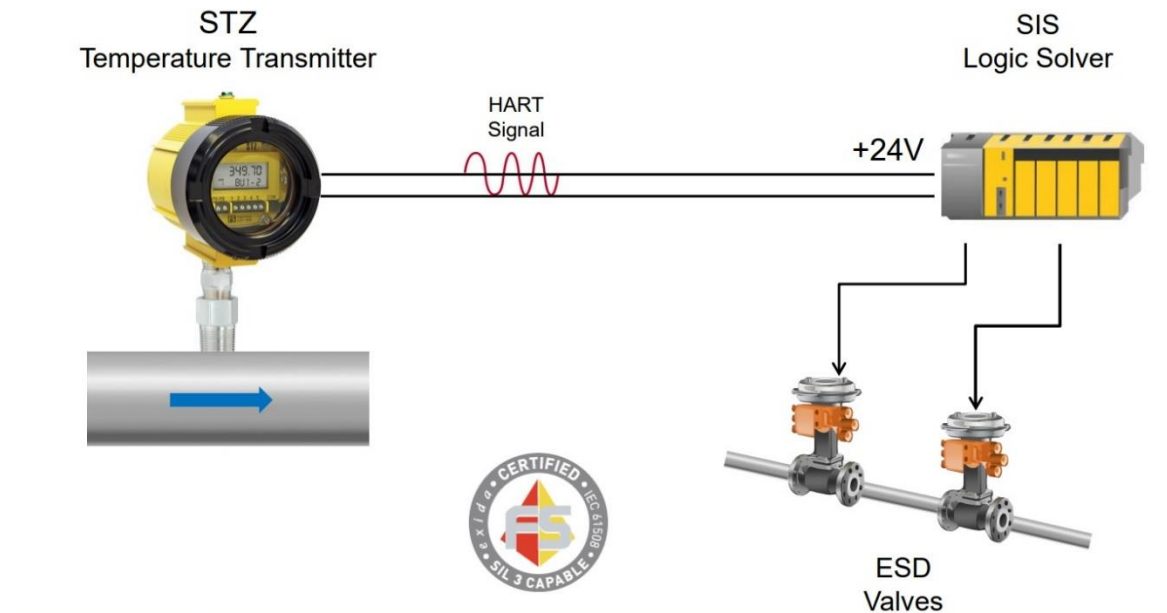
InstrumentationTools.com

ویژگی ها و امکانات ترانسمیتر دما

- ☐ نوع ورودی ها (ترموکوپل، PT100، انواع RTD ، ترمیستور)
- ☐ تعداد ورودی های قابل پذیرش
- ☐ شیوه اندازه گیری (دو سیمه، سه سیمه و ...)
- ☐ امکان ایزولاسیون، فیلتر کردن، خطی سازی درونی
- ☐ جبران سازی حرارتی برای ترموکوپل
- ☐ قابل نمایش دما و کنترل محلی
- ☐ نوع خروجی ها (ولتاژی، استانداردهای جریانی، شبکه های استاندارد صنعتی)
- ☐ محدوده مجاز کاری برای خود ترانسمیتر
- ☐ استانداردهای حفاظتی IP برای ترانسمیتر حفاظت در برابر ورود جامدات و مایعات (پاشش آب و غبار و ..)
- ☐ ایزولاسیون درونی (بریر داخلی)

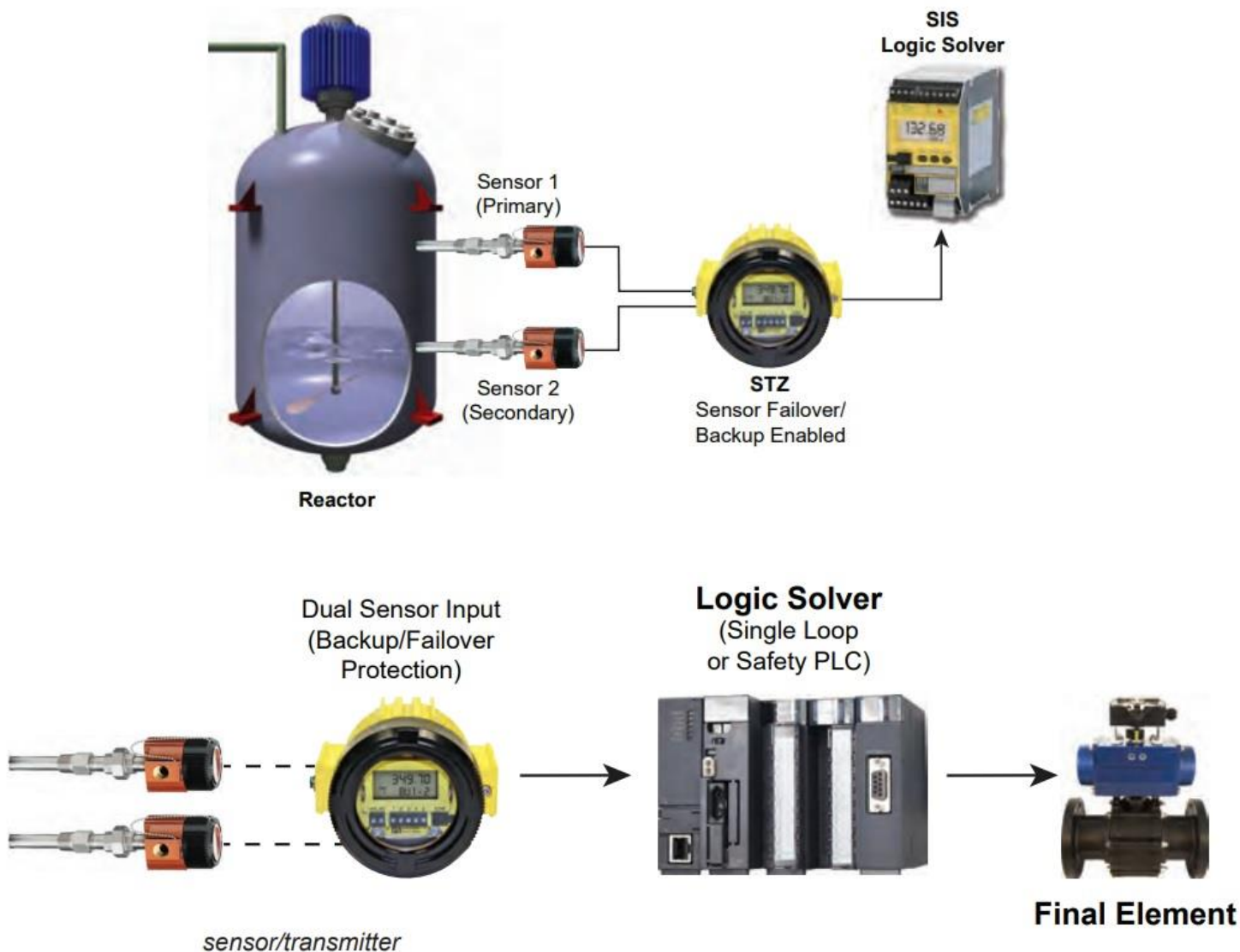
Temperature transmitters feature significant advances over their direct wired counterparts. They **eliminate special cabling requirements, simplify engineering and maintenance, as well as enabling advanced diagnostics**. The disadvantage of temperature transmitters is that they can obviously add extra **cost** to the temperature measurement system.

<https://www.bmengineering.co.uk/wp-content/uploads/2018/06/Temperature-Transmitter-Working-Principle.pdf>



<https://www.miinet.com/power-generation-transmission/sil-3-capable-temperature-transmitter>

مثالهایی از کاربرد ترانسمیتر دما در کنترل

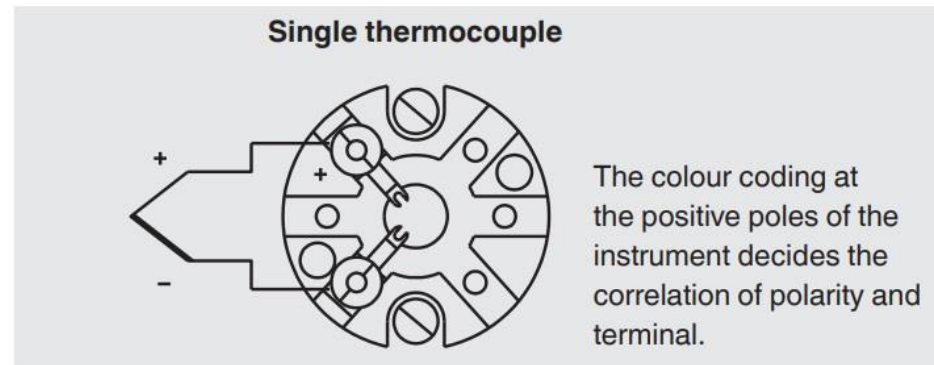


نمونه ای از یک ترموکوپل صنعتی و ترنسمیتر پیشنهادی شرکت WiKka



Special features

- Sensor ranges from -40 ... +1,200 °C [-40 ... +2,192 °F]
- With integrated fabricated protection tube
- Spring-loaded measuring insert (replaceable)
- Explosion-protected versions are available for many approval types (see page 2)



Connection head BSZ-H with LED display model DIH10
see data sheet AC 80.11

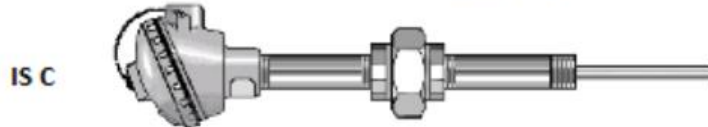
INDUSTRIAL SENSORS



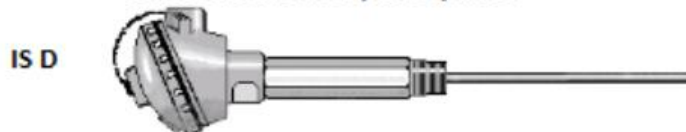
SLIDING COMPRESSION FITTING



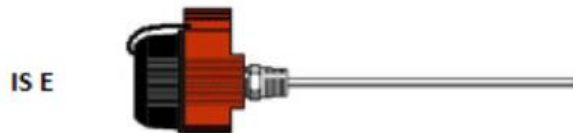
HEX NIPPLE



SPRING LOADED NIPPLE/UNION/NIPPLE



SPRING LOADED ADAPTOR



GENERAL PURPOSE SENSORS



SENSOR WITH EXTENSION LEAD



SENSOR WITH TAILS



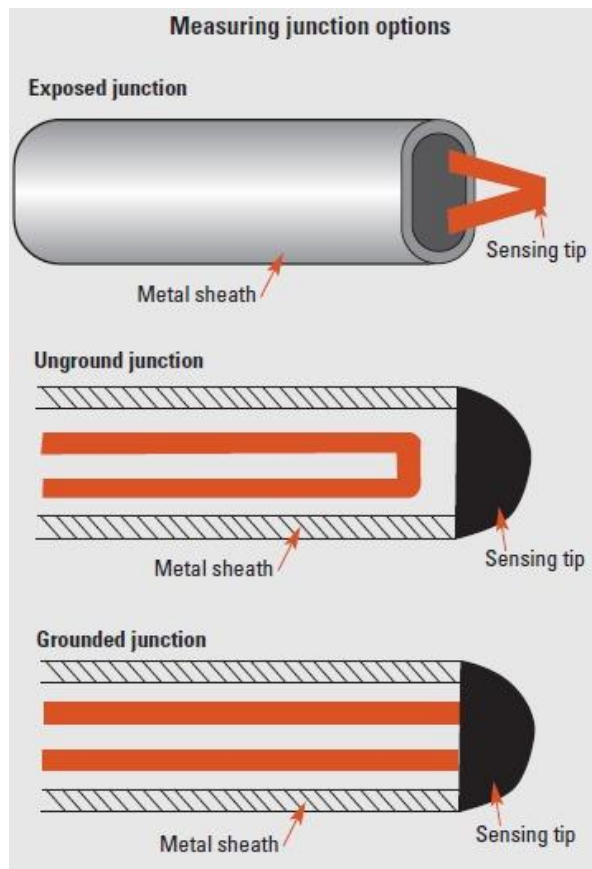
SENSOR WITH PLUG



SENSOR WITH HEX NIPPLE



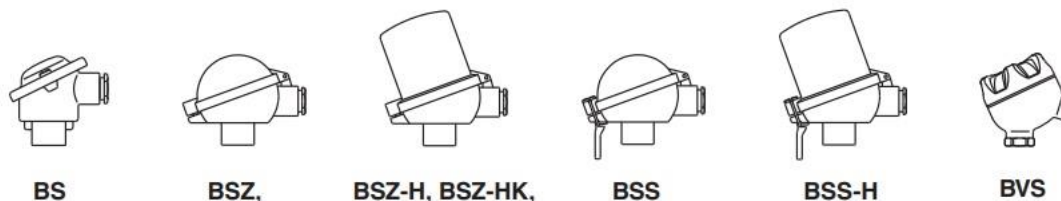
PAD TYPE SENSOR



<https://www.designworldonline.com/thermocouple-types/>

Connection head

■ European designs per EN 50446 / DIN 43735



Model	Material	Cable entry thread size	Ingress protection (max.) ¹⁾ IEC/EN 60529	Cap	Surface
BS	Aluminium	M20 x 1.5 or ½ NPT ³⁾	IP65 ⁴⁾	Flat cap with 2 screws	Blue, painted ⁵⁾
BSZ	Aluminium	M20 x 1.5 or ½ NPT ³⁾	IP65 ⁴⁾	Spherical hinged cover with cylinder head screw	Blue, painted ⁵⁾
BSZ-H	Aluminium	M20 x 1.5 or ½ NPT ³⁾	IP65 ⁴⁾	Raised hinged cover with cylinder head screw	Blue, painted ⁵⁾
BSZ-H / DIH10 ²⁾	Aluminium	M20 x 1.5 or ½ NPT ³⁾	IP65	Raised hinged cover with cylinder head screw	Blue, painted ⁵⁾
BSS	Aluminium	M20 x 1.5 or ½ NPT ³⁾	IP65	Spherical hinged cover with clamping lever	Blue, painted ⁵⁾
BSS-H	Aluminium	M20 x 1.5 or ½ NPT ³⁾	IP65	Raised hinged cover with clamping lever	Blue, painted ⁵⁾
BVS	Stainless steel	M20 x 1.5 ³⁾	IP65	Precision-cast screw-on lid	Blank, electropolished
BSZ-K	Plastic	M20 x 1.5 or ½ NPT ³⁾	IP65	Spherical hinged cover with cylinder head screw	Black
BSZ-HK	Plastic	M20 x 1.5 or ½ NPT ³⁾	IP65	Raised hinged cover with cylinder head screw	Black

WIKA data sheet TE 65.06 · 05/2021

What is Ingress Protection (IP)?

Ingress Protection (IP) codes are defined by the [International Electrotechnical Commission](https://www.electrotechnicalcommission.org/) and are a combination of two digits representing the device's level of protection from the ingress of solids 1st digit and liquids 2nd digit.

The first digit ranges from 0 no protection to 6 dust resistance. The second digit ranges from 0 no protection to 9k protection against high pressure, high temperature water—such as in wash down applications

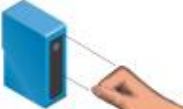
2nd Digit = Protection Against Ingress of Water



IP Codes:

2nd Digit = Protection Against Ingress of Water

1st Digit = Protection Against Ingress of Foreign Bodies

IP 0 ... No protection	
IP 1 ... Size of foreign body ≥ 50 mm	
IP 2 ... Size of foreign body ≥ 12 mm	
IP 3 ... Size of foreign body ≥ 2.5 mm	
IP 4 ... Size of foreign body ≥ 1 mm	
IP 5 ... Dust protected	
IP 6 ... Dust proof	

										
	IP ...0 No protection	IP ...1 Drip-water vertical	IP ...2 Drip-water tilted	IP ...3 Spray water	IP ...4 Splash water	IP ...5 Jet water	IP ...6 Strong jet of water	IP ...7 Temporary immersion	IP ...8 Lasting immersion	IP ...9K 100 bar, 16 l/min., 80° C
IP 00										
IP 10		IP 11	IP 12							
IP 20		IP 21	IP 22	IP 23						
IP 30		IP 31	IP 32	IP 33	IP 34					
IP 40		IP 41	IP 42	IP 43	IP 44					
IP 50				IP 53	IP 54	IP 55	IP 56			
IP 60						IP 65	IP 66	IP 67		IP 69K

What are Safety Integrity Levels?

Safety Integrity Level, or as more usually referred to "SIL" or "SIL level", is a unit of measurement for quantifying risk reduction.

There are four integrity levels. Each of the 4 levels of SIL represents an order of magnitude of risk reduction.

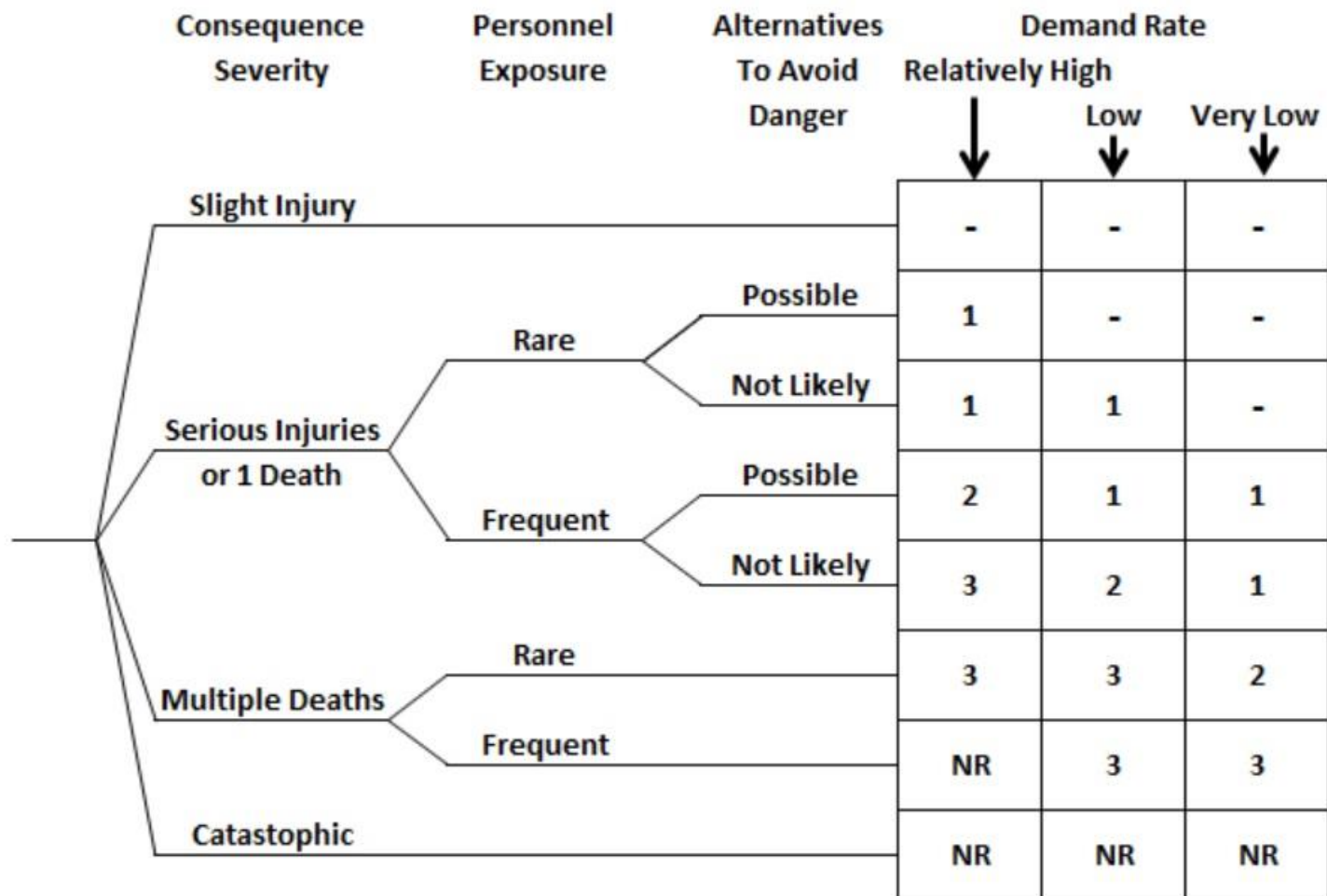
SIL 1 - represents the integrity required to avoid relatively minor incidents and is likely to be satisfied by a certain degree of fault tolerant design using guidelines that follow good practice.

SIL 2 - represents the integrity to avoid more serious, but limited, incidents some of which may result in serious injury or death to one or more persons.

SIL 3 - represents the integrity required to avoid serious incidents involving a number of fatalities and/or serious injuries.

SIL 4 - represents the integrity level required to avoid disastrous accidents

Determining SIL Level (SIL Rating)- SIL Tree



- = No special safety feature required
 NR = Not recommended, consider alternatives

Safety Integrity Level (SIL)

www.Control and Instrumentation.com

Hazardous Area?

HAZARDOUS AREAS GAS GROUPS

Products manufactured for safe installation in explosive atmospheres and hazardous area zones (other than mines susceptible to firedamp) and categorized according to ATEX as follows:

Zone 0 Group II Category 1G – the presence of explosive atmospheres is continuous, frequent or for long time periods

Zone 1 Group II Category 2G – the presence of explosive atmospheres is intermittent [not steady] in normal operation (likely)

Zone 2 Group II Category 3G — the presence of explosive atmospheres is occasional or for short periods (never in normal operation)

The [ATEX directives](#) are a pair of directives from the European Committee for Standardization created to protect people from ATmosphères EXplosibles—explosive atmospheres—in the European Union.

What is Barrier?

Barriers

Intrinsic devices are that which generate a certain amount of thermal and electrical energy which can cause fire or explosion in [hazardous areas](#).

Mostly, in hazardous areas where there are high chances of ignition or other forms of damage, you have to take special care to protect the [field instruments](#) from damaging themselves or the instrumentation devices inside the panel.

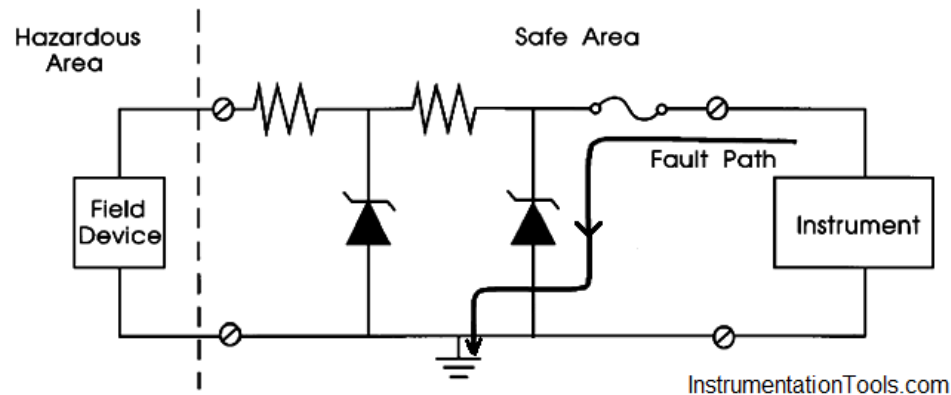
<https://instrumentationtools.com/use-of-isolators-and-barriers-in-electrical-panel/>

One method of protection is to use temperature sensor Zener Diode barriers which are simple passive devices comprised of Zener diodes, resistors and fuses that serve to limit the voltage, current, and power available to the hazardous area sensor or device. Another protection method is to install **Isolated barriers** for your temperature sensors located in hazardous areas. These barriers require auxiliary operating power and cost more than passive **Zener barriers**.

<https://www.miinet.com/power-generation-transmission/sil-3-capable-temperature-transmitter>

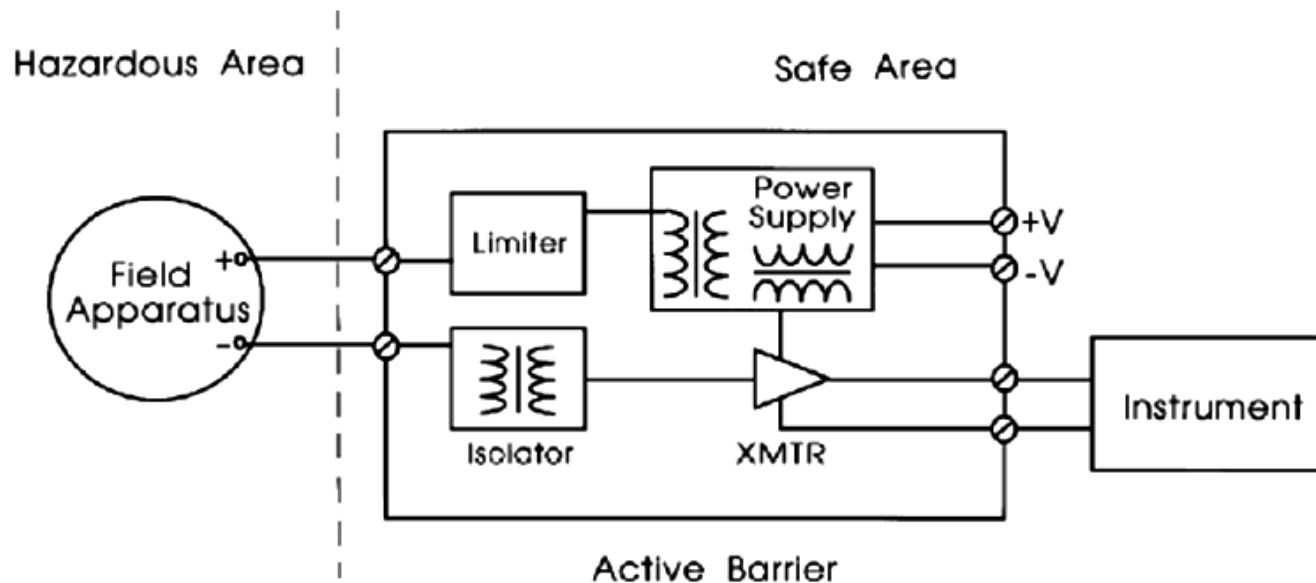
Barrier Types:

Zener diode barrier



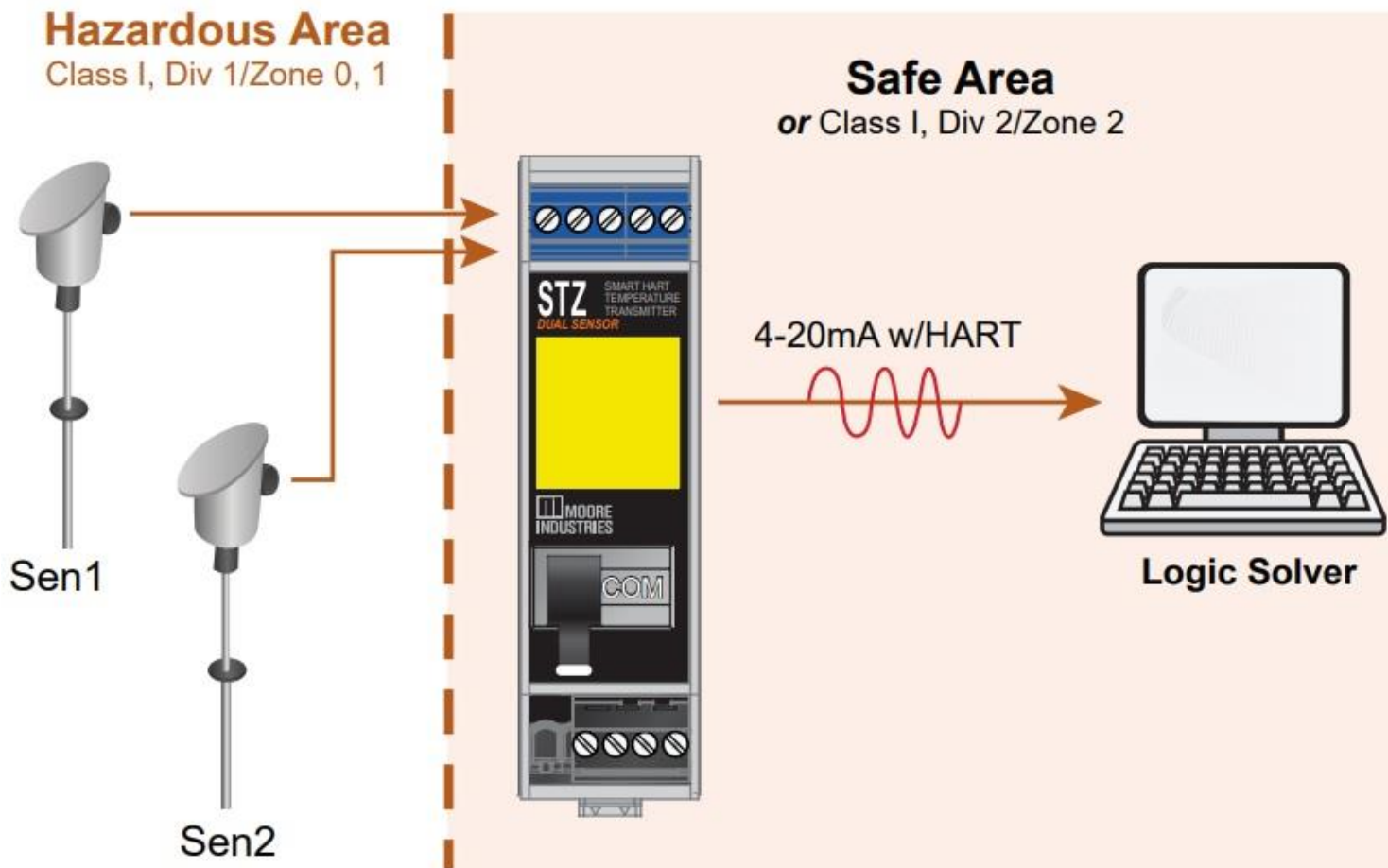
<https://instrumentationtools.com/zener-diode-barrier-principle/>

Active barrier



<https://instrumentationtools.com/what-is-a-active-barrier/>

استفاده از یک ترنسیمیتر با بریر داخلی



انواع شکلهای ظاهری ترانسمیترهای شرکت Moore



STZ in HPP
Encapsulated
Housing



STZ in SB
316 Stainless Steel
Field-Mount Enclosure



STZ in LH2
Aluminum Connection
Head Field-Mount
Enclosure



STZ in DIN
Rail Mount Housing



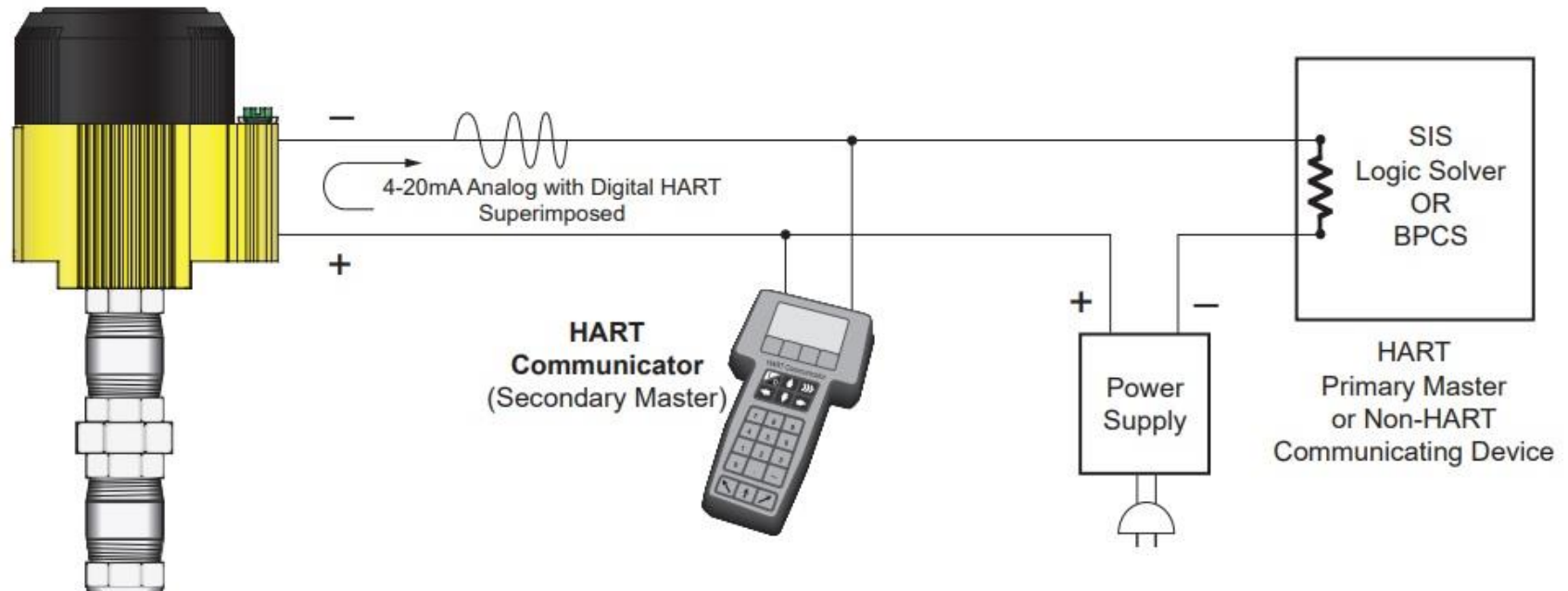
STZ in BH
Aluminum Field-Mount
Enclosure



STZ in HP
Hockey-Puck Housing
with Display

ترانسمیتر با خروجی آنالوگ جریان و HART

Functional Safety
Smart HART
Temperature Transmitter

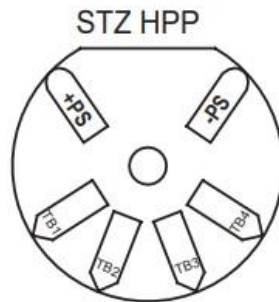


Unit	Input	Output	Power
STZ Functional Safety Dual Input Smart HART Temperature Transmitter	PRG Programmable with standard HART RTD 2-, 3-, 4-Wire Platinum, Copper, Nickel Thermocouple (J, K, E, T, R, S, B, N, C) 0-4000 ohms -50-1000mV	4-20MA Scalable to narrower ranges	12-42DC* 12-30DC Intrinsically-Safe (I.S.) applications * Non-incendive (Class 1 Div 2, Zone 2) by default

STZ in HPP
Encapsulated
Housing



STZ HPP (4 Terminals) Input Connections



Note: STZ [HPP] (see Figure 7) – When using two input sensors, Sensor 1 can be configured as 2-wire or 3-wire sensor. Sensor 2 is then restricted to a 2-wire sensor.

